MOCK TEST PAPER 3

General Instructions: *Same as Mock Test Paper 1.*

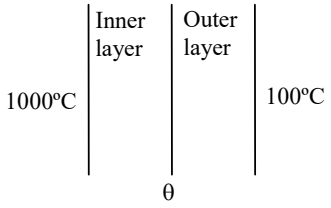
Physics

Section A

Q. 1. If 50 Vernier divisions are equal to 49 main scale divisions of a travelling microscope and the smallest reading of the main scale is 0.5 mm, the Vernier constant of travelling microscope is:
(1) 0.1 mm **(2)** 0.1 cm **(3)** 0.01 cm **(4)** 0.01 mm

Q. 2. A vessel is filled with a gas at a pressure of 76 cm of mercury at a certain temperature. The mass of the gas is increased by 50% by introducing more gas into the vessel at the same temperature. The resultant pressure, in cm of Hg, is:
(1) 76 **(2)** 152 **(3)** 114 **(4)** 1117

Q. 3. The temperature drop through a two-layer furnace wall is 900°C. Each layer is of equal area of cross-section. Which of the following actions will result in a decrease in the temperature θ of the interface?

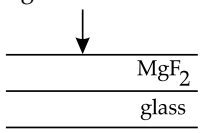


(1) By increasing the thermal conductivity of the outer layer
(2) By increasing the thermal conductivity of the inner layer
(3) By increasing thickness of outer layer
(4) By decreasing thickness of inner layer

Q. 4. In a resonance pipe, the first and second resonances are obtained at lengths 22.7 cm and 70.2 cm, respectively. Which of the following is an appropriate estimate of the end correction of the pipe?
(1) 1.05 cm **(2)** 115.5 mm
(3) 92.5 mm **(4)** 2.135 cm

Q. 5. A long solenoid has 1000 turns. When a current of 4 A flows through it, the magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. The self-inductance of the solenoid is:
(1) 4 H **(2)** 3 H **(3)** 2 H **(4)** 1 H

Q. 6. White light is incident normally on a glass surface (n = 1.52) coated with a film of MgF_2 (n = 1.38). At what minimum thickness of the film will yellow light of wavelength 550 nm (in air) be missing in the reflected light?



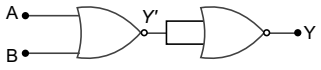
(1) 99.6 nm **(2)** 49.8 nm
(3) 19.6 nm **(4)** 10.6 nm

Q. 7. A uniform magnetic field of 2×10^{-3} T acts along a positive Y-direction. A rectangular loop of sides 20 cm and 10 cm with a current of 5A is in the Y-Z plane. The current is in an anticlockwise sense with reference to the negative X axis. Magnitude and direction of the torque is:
(1) 2×10^{-4} N – m along positive Y-direction
(2) 2×10^{-4} N – m along positive Z-direction
(3) 2×10^{-4} N – m along positive X-direction
(4) 2×10^{-4} N – m along negative Z-direction

Q. 8. A mass m is revolving in a vertical circle at the end of a string of length 20 cm. By how much does the tension of the string at the lowest point exceed the tension at the top most point?
(1) 2 mg **(2)** 4 mg **(3)** 6 mg **(4)** 8 mg

Q. 9. In Young's experiment, the ratio of maximum and minimum intensities in the fringe system is 9:1. The ratio of amplitudes of coherent sources is:
(1) 9:1 **(2)** 3:1 **(3)** 2:1 **(4)** 1:1

Q. 10. In the following circuit, the output Y for all possible inputs A and B is expressed by the truth table:



(1)

A	B	Y
0	1	1
0	1	1
1	0	1
1	1	0

(2)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(3)

A	B	Y
0	1	1
1	0	1
0	0	0
1	1	1

(4)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Q. 11. Moment of inertia of a uniform circular disc about a diameter is I. Its moment of inertia about an axis perpendicular to its plane and passing through a point on its rim will be:
(1) 5I **(2)** 3I **(3)** 6I **(4)** 4I

Q. 12. Given below are two statements:

Statement I: The ratio of the inertial mass to gravitational mass is equal to 1.

Statement II: The Equivalence Principle states that inertial mass and gravitational mass may differ significantly in certain situations.

Choose the correct answer from the options given below:

(1) Statement I is true, Statement II is true and Statement II is the correct explanation of Statement I.

(2) Statement I is true, Statement II is true, but Statement II is not the correct explanation of Statement I.

(3) Statement I is true, Statement II is false.

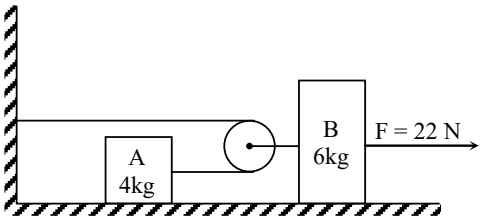
(4) Statement I is false, Statement II is true.

Q. 13. Stopping potentials of 24, 100, 110 and 115 kV are measured for photoelectrons emitted from a certain element when it is irradiated with monochromatic X-rays. If the element is used as a target in an X-ray tube, the energy of Kα line is:
(1) 54 KeV **(2)** 76 KeV
(3) 88 KeV **(4)** 32 KeV

Q. 14. A heavy nucleus with mass number 200 gets disintegrated into two small fragments with mass number 80 and 120. If binding energy per nucleon for the parent atom is 6.5 MeV and for daughter nuclei is 7 MeV and 8 MeV, respectively, then the energy released in the decay will be:
(1) 200 MeV **(2)** –220 MeV
(3) 220 MeV **(4)** 180 MeV

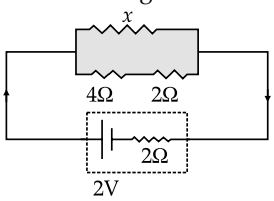
Q. 15. A particle of mass 'm' is executing oscillations about the origin on the X-axis. Its potential energy is $U(x) = k|x|^3$ where k is a positive constant. If the amplitude of oscillation is 'a', then its time period T is:
(1) Proportional to $\frac{1}{\sqrt{a}}$ **(2)** Independent of a
(3) Proportional to $\sqrt{a}$ **(4)** Proportional to $a^{3/2}$

Q. 16. Two blocks are connected by a massless string through an ideal pulley, as shown. A force of 22 N is applied on block B when the blocks are at rest. The acceleration of the centre of mass of block A and block B, 2 s after the application of force is (masses of A and B are 4 kg and 6 kg, respectively and surfaces are smooth):



(1) 1.4 m/s^2 **(2)** 1 m/s^2
(3) 2 m/s^2 **(4)** None of these

Q. 17. Observe the following circuit.



In order to ensure that the power dissipated in external circuit is maximum, the value of x should be (in ohm) _____ .
(1) 4Ω **(2)** 3Ω **(3)** 2Ω **(4)** 1Ω

Q. 18. In a series LCR circuit, voltage drop across resistance is 8 V; across inductor is 6 V and across capacitor is 12 V. Then:

(1) Voltage of the source will be leading current in the circuit.

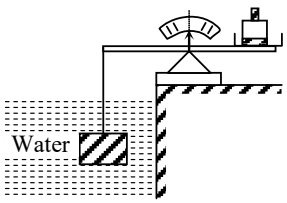
(2) Voltage drop across each element will be less than the applied voltage.

(3) Power factor of the circuit will be $\frac{4}{3}$.

(4) None of these

Q. 19. If an electron revolves around a proton, then its time period (R = radius of orbit):
(1) T ∝ R^2 **(2)** T ∝ R$^{3/2}$
(3) T ∝ R^3 **(4)** T ∝ R

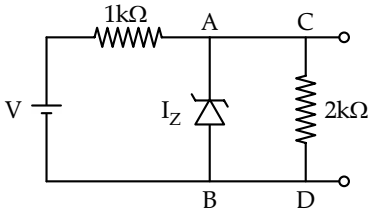
Q. 20.



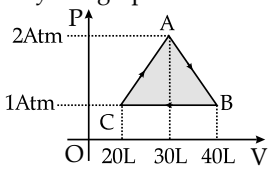
The volume of a brick is 2.197 litres. The submerged brick is balanced by a 2.54 kg mass on the beam scale. The weight of the brick is: (g = 9.8 m/s^2)
(1) 46 N **(2)** 50 N **(3)** 56 N **(4)** 72 N

Section B

Q. 21. In the given circuit, the breakdown voltage of Zener diode is 3.0 V. The value of Iz will be .....mA.

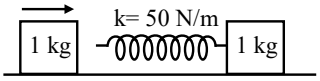


Q. 22. Consider a PV–cyclic process ABCA as per described by the graph below.



Heat of 1205.0 J is provided for the process to take place. What is the amount of heat rejected (in joule) in one cycle?

Q. 23. Each of the blocks shown in the figure has mass 1 kg. The rear block moves with a speed of 2 m/s towards the front block kept at rest. The spring attached to the front block is light and has a spring constant 50 N/m. The maximum compression of the spring is given by $\frac{X}{10}$ m, then the value of X is ..... m.



Q. 24. In a certain region of space, electric potential V is given by $V = ax^2 + ay^2 + 2az^2$ (where 'a' is a constant of proper dimensions). Work done by electric field in moving a 2μC charge from origin to (0, 0, 0.1 m) is (-5×10^{-8}) J. The approximate value of 'a' in V/m^2 is ..... .

Q. 25. A wire of length 2m is clamped horizontally between two fixed supports. A mass m = 5 kg is hung from the middle of the wire. The vertical depression in wire in equilibrium is ..... cm. (Young modulus of wire = 2.4×10^9 N/m^2, cross-sectional area = 1 cm^2)

Chemistry

Section A

Q. 26. For a molecule Br_2, total distance between two nuclei is 3.2 Å. What will be the covalent radius of Br atom?

(1) 1.6 Å **(2)** 6.4 Å **(3)** 2.4 Å **(4)** 4.9 Å

Q. 27. Match the items under list (1) with items under list (2) and select the correct answers from the sets (1), (2), (3) and (4):

List (1) Molecule	List (2) Shape
(a) PCl_5	**(i)** V-shaped
(b) F_2O	**(ii)** Triangular planar
(c) BCl_3	**(iii)** Trigonal bipyramidal
(d) NH_3	**(iv)** Trigonal pyramidal
	(v) Tetrahedral

(1) a-i, b-v, c-iv, d-iii **(2)** a-ii, b -iii, c-i , d-ii
(3) a-iv, b-iii, c-ii, d-v **(4)** a-iii, b-i, c-ii, d-iv

Q. 28. During the qualitative analysis of a salt with cation y^{2+}, the addition of a reagent (X) to an alkaline solution of the salt gives a bright red precipitate. The reagent (X) and the cation (y^{2+}) present respectively are:

(1) Dimethylglyoxime and Ni^{2+}
(2) Dimethylglyoxime and Co^{2+}
(3) Nessler's reagent and Hg^{2+}
(4) Nessler's reagent and Ni^{2+}

Q. 29. Calculate the work done when 2 moles of hydrogen expand isothermally and reversibly at 27°C from 15 to 50 litres.

(1) 14.45 k cal **(2)** 1445 J
(3) –1445 cal **(4)** 14.45 kJ

Q. 30. The k_{sp} for bismuth sulphide (Bi_2S_3) is 1.08×10^{-73}. The solubility of Bi_2S_3 in mol/L at 198 K is

(1) 2.7×10^{-12} **(2)** 1.0×10^{-15}
(3) 4.2×10^{-8} **(4)** 3.2×10^{-10}

Q. 31. Calculate the pH of 0.1 M CH_3COOH

(Given $K_{a_{CH_3COOH}} = 2 \times 10^{-5}$)

(1) 2.5 **(2)** 2.2
(3) 2.85 **(4)** 3.15

Q. 32. IUPAC name of:

$$H_3C - \underset{\displaystyle C_2H_5}{\underset{|}{CH}} - CH_2 - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_2Cl \text{ is:}$$

(1) 1-chloro-4-methylhexan-2-ol
(2) 1-chloro-4-methylhexan-2-al
(3) 1-chloro-4-ethylpentan-2-ol
(4) 1-chloro-2-hydroxy-4-methylhexane

Q. 33. Given below are two statements:

Statement I: The number of geometrical isomers in asymmetrical molecule are 2^n.

Statement II: To show geometrical isomerism, molecule must be plannar and have restricted rotation.

(1) Both Statement I and Statement II are true.
(2) Both Statement I and Statement II are false.
(3) Statement I is true but Statement II is false.
(4) Statement I is false but Statement II is true.

Q. 34. Given below are two statements.

Statement I: The presence of weaker π–bonds make alkenes less stable than alkanes.

Statement II: The strength of the double bond is greater than that of the carbon-carbon single bond.

In the light of the above statements, choose the *correct* answer from the options given below.

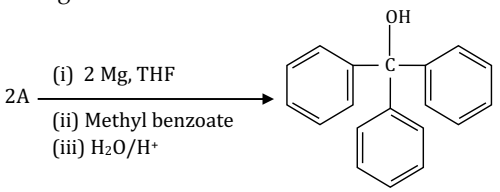
(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

Q. 35. In the given reaction



(1) Benzyl bromide **(2)** bromobenzene

(3) cyclohexyl bromide **(4)** methyl bromide

Q. 36. Cerium (IV) has a noble gas configuration. Which of the following is correct statement about it?

(1) It will not prefer to undergo redox reactions.

(2) It will prefer to gain electron and act as an oxidising agent.

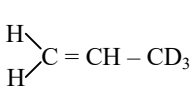
(3) It will prefer to give away an electron and behave as reducing agent.

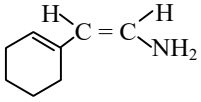
(4) It acts as both, oxidising and reducing agent.

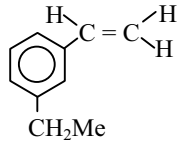
Q. 37. The depression in freezing point observed for a formic acid solution of concentration 0.5 mol L^{-1} is 0.0405°C. The density of formic acid is 1.05 g mL^{-1}. The Van't Hoff factor of the formic acid solution is nearly: (Given for water, k_f = 1.86 K kg mol^{-1})

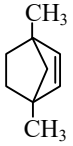
(1) 0.8 **(2)** 1.1 **(3)** 1.9 **(4)** 2.4

Q. 38. Hyperconjugation is possible in:

(i) 

(ii) 

(iii) 

(iv) 

(1) i and ii **(2)** i, ii and iii

(3) only ii **(4)** All of the above

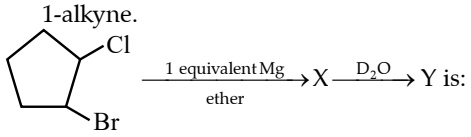
Q. 39. When 1-alkyne is treated with Na + Liq. NH_3 and product is reacted with methyl chloride, the end product of the reaction will be:

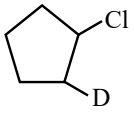
(1) lower alkyne having two carbon less than 1-alkyne.

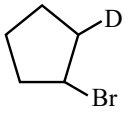
(2) lower alkyne having one carbon less than 1-alkyne.

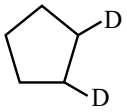
(3) higher alkyne having one carbon more than 1-alkyne.

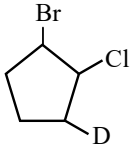
(4) higher alkyne having two carbon more than 1-alkyne.

Q. 40. 

(1) 

(2) 

(3) 

(4) 

Q. 41. In 'nitration mixture' concentrated sulphuric acid is used:

(1) as sulphonating agent.

(2) as dehydrating agent.

(3) for the formation of nitronium ions.

(4) as a solvent.

Q. 42. Methanol and ethanol are distinguished by:

(1) Treating with Victor Meyer test

(2) Treating with Lucas reagent

(3) Heating with iodine and alkali

(4) Treating with CrO_3 in dil. H_2SO_4

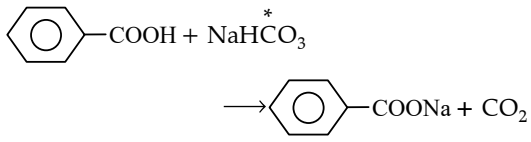
Q. 43. In which of the following crossed aldol condensations, only one kind of cross aldol is formed?

(1) CH_3CHO and CH_3CH_2CHO

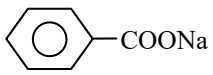
(2) CH_3CHO and $(CH_3)_2CO$

(3) $(CH_3)_2CO$ and $(C_2H_5)_2CO$

(4) C_6H_5CHO and CH_3CHO

Q. 44. 

C* is with in the product—

(1) CO_2 **(2)** 

(3) Both (1) and (2) **(4)** None of these

Q. 45. Name the products in the acid- base reaction:

(A) $CH_3CH_2NH_2$ + HI

(B) $(CH_3)_3N$ + HBr

(1) **(A)** Trimethyl ammonium iodide
(B) Trimethyl ammonium bromide

(2) **(A)** Ethyl ammonium iodide
(B) Methyl ammonium bromide

(3) **(A)** Ethyl ammonium iodide
(B) Trimethyl ammonium bromide

(4) All of the above

Section B

Q. 46. Two substances, A ($t_{1/2}$ = 5 min) and B ($t_{1/2}$ = 15 min) are taken in such a way that initially [A] = 4[B]. The time after which both the concentration will be equal is ..... (assuming reactions are of first order).

Q. 47. The standard free energy change for the reaction:

$H_2(g) + 2AgCl(s) \rightarrow 2Ag_{(s)} + 2H^+_{(aq)} + 2Cl^-_{(aq)}$

is –10.26 kcal mol^{-1}at 25°C. A cell using above reaction is operated at 25°C under P_{H_2} = 1 atm, $[H^+]$ and $[Cl^-]$ = 0.1. The e.m.f. of cell is ..... V.

Q. 48. In the given sequence, identify the no. of *c*-atom present in the final product (D).

$$CH_3CH_2OH \xrightarrow{[O]} A \xrightarrow[NaOH]{I_2} B + C$$

$$C \xrightarrow{\text{Silver powder}} D$$

Q. 49. For the reaction

$H_2F_2(g) \rightarrow H_2(g) + F_2(g)$,

$\Delta U = -59.6$ kJ mol^{-1} at 27 °C.

The enthalpy change for the above reaction is (–) _____ kJ mol^{-1} [nearest integer]

Given: R = 8.314 J K^{-1} mol^{-1}.

Q. 50. **(a)** The number of unpaired e^- in $[FeF_6]^{3-}$ is *x*.

(b) The number of geometrical isomers in $[Pt\ Cl\ Br\ IF]^{2-}$ is *y*.

(c) The oxidation state of central atom in $Ni(CO)_4$ is Z.

Find the sum value of *x*, *y* and *z*.

Mathematics

Section A

Q. 51. The minimum number of elements that must be added to the relation R = {(*a*, *b*), (*b*, *c*)} on the set {*a*, *b*, *c*}, so that it becomes symmetric and transitive is:

(1) 3 **(2)** 7 **(3)** 5 **(4)** 4

Q. 52. If D, E, F are the mid points of the sides BC, CA and AB respectively of a triangle ABC and 'O' is any point, then, $|\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF}|$, is:

(1) 1 **(2)** 0 **(3)** 2 **(4)** 4

Q. 53. If $I_m = \int \frac{x^m}{y} dx$, $y^2 = 2x^2 + 4x + 3$ and constants *a*, *b*, *c* such that $aI_3 + bI_2 + cI_1 = x^2y$, then ${}^{a+c}P_{b+1}$ is equal to:

(1) 11! **(2)** 10! **(3)** 12! **(4)** 14!

Q. 54. $\int_0^\infty \left[\frac{2}{e^x}\right] dx$ is equal to, where [.] denotes greatest integer function.

(1) e^2 **(2)** $\log_e 2$ **(3)** 0 **(4)** $\frac{1}{e}$

Q. 55. Area bounded by the curves $x = |y^2 - 1|$ and $y = x - 5$ is equal to:

(1) $\frac{109}{6}$ sq. units **(2)** $\frac{121}{6}$ sq. units

(3) $\frac{113}{6}$ sq. units **(4)** $\frac{117}{6}$ sq. units

Q. 56. $\int \sqrt{1+x^{2n}} \left(\frac{\log_e(1+x^{2n}) - 2n\log_e x}{x^{3n+1}} \right) dx, n \in a$

equals to:

(1) $-\frac{2}{3n}\frac{\sqrt{(1+x^{2n})^3}}{x^{3n}}\left[\frac{1}{2}\log_e(1+x^{2n}) - \frac{1}{2}\log_e x^n + \frac{1}{3}\right] + c$

(2) $-\frac{2}{3n}\frac{\sqrt{(1+x^{2n})^3}}{x^{3n}}\left[\log_e(1+x^{2n}) - \log_e x^{2n} - \frac{1}{3}\right] + c$

(3) $-\frac{2}{3n}\frac{\sqrt{(1+x^{2n})^3}}{x^{3n}}\left[\frac{1}{2}\log_e(1+x^{2n}) - \frac{1}{2}\log_e x^{2n} - \frac{1}{3}\right] + c$

(4) $-\frac{2}{3n}\frac{\sqrt{(1+x^{2n})^3}}{x^{3n}}\left[\frac{1}{2}\log_e(1+x^{2n}) + \frac{1}{2}\log_e x^{2n} + \frac{1}{3}\right] + c$

Q. 57. If $y = a\log|x| + bx^2 + x$ has its extremum values at $x = -1$ and $x = 2$, then:

(1) $a = 2, b = -1$ **(2)** $a = 2, b = \frac{-1}{2}$

(3) $a = -2, b = \frac{1}{2}$ **(4)** a = –2, $b = \frac{-1}{2}$

Q. 58. If $x = a[\cos\theta + \log\tan\frac{\theta}{2}]$, $y = a\sin\theta$ then $\frac{dy}{dx}$ =

(1) $\cos\theta$ **(2)** $\sin\theta$ **(3)** $\tan\theta$ **(4)** $\operatorname{cosec}\theta$

Q. 59. If the range of the function $y = \log_{\sqrt{2}}(2 - \log_2(16\sin^2 x + 1))$ is (*q*, P] then the value of P is:

(1) $\sqrt{2}$ **(2)** 4 **(3)** 2 **(4)** ∞

Q. 60. The smallest value of k^2 for which $\lim_{x\to 1} \sec^{-1}\left(\frac{k^2}{\log_e x} - \frac{k^2}{x-1}\right)$ exists:

(1) 1 **(2)** 2 **(3)** 4 **(4)** 3

Q. 61. A circle touching the x-axis at (3, 0) and making an intercept of length 8 on the y-axis passes through the point:

(1) (3, 10) **(2)** (3, 5)

(3) (2, 3) **(4)** (1, 5)

Q. 62. **Statement 1:** If $\sum_{i=1}^{8}(x_i-7) = 8$ and $\sum_{i=1}^{8}(x_i-7)^2 = 40$, then standard deviation of $x_1, x_2, x_3, ..., x_8$ is 2.

Statement 2: Standard deviation is independent of change of origin.

(1) Statement 1 is true, Statement 2 is true and Statement 2 is correct explanation for Statement-1.

(2) Statement 1 is true, Statement 2 is true and Statement 2 is not correct explanation for Statement 1.

(3) Statement 1 is true, Statement 2 is false.

(4) Statement 1 is false, Statement 2 is true.

Q. 63. Let a perpendicular AB be drawn from A (1, 6, 3) to the line $\frac{x}{1} = \frac{y-1}{1} = \frac{z-2}{3}$, where B is foot of perpendicular, then:

(1) B is (2, 3, 5)

(2) Image of point A *w.r.t.* given line is (1, 0, 4)

(3) Projection of the line segment AB on a line having direction cosines $\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}$ is $\sqrt{2}$.

(4) AB = $\sqrt{12}$

Q. 64. Let the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{7}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{\alpha}=\frac{1}{25}$ coincide. Then the length of the latus rectum of the hyperbola is:

(1) $\frac{32}{9}$ **(2)** $\frac{18}{5}$ **(3)** $\frac{27}{4}$ **(4)** $\frac{27}{10}$

Q. 65. For the roots of the equation $a - bx - x^2 = 0$; ($a > 0, b > 0$), which statement is true?

(1) both roots are positive

(2) both roots are negative

(3) roots have opposite sign, negative root has greater magnitude

(4) roots have opposite sign, positive root has greater magnitude

Q. 66. Consider the system of linear equations:

$$-x + y + 2z = 0$$
$$3x - ay + 5z = 1$$
$$2x - 2y - az = 7$$

Let P be the set of all $a \in R$ for which the system is inconsistent and Q be the set of all $a \in R$ for which the system has infinitely many solutions. If ΣP and ΣQ denotes the sum of all elements of set P and set Q respectively, then $\Sigma P + \Sigma Q$ equals to:

(1) 2 **(2)** 7 **(3)** 10 **(4)** 14

Q. 67. If the ratio of the sum of n terms of two AP's is $2n : (n+1)$, then ratio of their 8th terms is

(1) 15 : 8 **(2)** 8 : 13 **(3)** 11 : 6 **(4)** 5 : 17

Q. 68. If the 4th term in the expansion of $\left(ax+\frac{1}{x}\right)^n$ is $\frac{5}{2}$, then the values of a and n respectively are:

(1) 2, 6 **(2)** $\frac{1}{2}$, 6

(3) $\frac{1}{2}$, 5 **(4)** 2, 5

Q. 69. The number of positive integers satisfying the inequality ${}^{n+1}C_{n-2} - {}^{n+1}C_{n-1} \le 100$ is:

(1) Nine **(2)** Eight **(3)** Five **(4)** Ten

Q. 70. If the orthocentre of the triangle formed by (1, 3) (4, –5) and (a, b) is (2, 4), Then the value of $33b + 22a$ is:

(1) 0 **(2)** $\frac{1}{11}$ **(3)** 1 **(4)** $\frac{3}{11}$

Section B

Q. 71. A line with direction ratios <0, 2, –1> meets the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$ and $\frac{x-1}{1} = \frac{y+2}{3} = \frac{z-2}{-2}$ at P and Q respectively. If area of triangle OPQ, where O is origin, is equal to $\frac{1}{n}\sqrt{m}$, then ($m - n$) equals to ... .

Q. 72. In a workshop, there are five machines and the probability of any one of them to be out of service on a day is $\frac{1}{4}$. If the probability that at most two machines will be out of service on the same day is $\left(\frac{3}{4}\right)^3 k$, then gk is equal to.....

Q. 73. If $m = \int_{-1}^{10} \text{sgn}(x-[x])dx$

and $n = \int_0^{26} \frac{e^{\cos\pi\{x\}}}{e^{\cos\pi\{x\}}+e^{-\cos\pi\{x\}}}dx$, then ${}^nc_m$ equals to; where [.] and {.} denote greatest integer function & fraction part function respectively.

Q. 74. If $D=\begin{bmatrix} 0 & a\alpha^2 & a\beta^2 \\ b\alpha+c & 0 & a\gamma^2 \\ b\beta+c & (b\gamma+c) & 0 \end{bmatrix}$ is a skew symmetric matrix (where α, β, γ are distinct) and the value of $\begin{vmatrix} (a+1)^2 & (1-a) & (2-c) \\ (3+c) & (b+2)^2 & (b+1)^2 \\ (3-b)^2 & b^2 & (c+3) \end{vmatrix}$ is λ, then the value of $|10\lambda|$ is.....

Q. 75. Let $|z| = |z-3| = |z-4i|$, then the value $|2z|$ is.....

Answers

Physics

Q. No.	Answer	Topic's name	Q. No.	Answer	Topic's name
1	(3)	Experimental Physics	14	(3)	Nuclear Fission
2	(3)	Kinetic Theory of Gases	15	(1)	Unit and Measurement
3	(1)	Heat Transfer	16	(1)	Laws of Motion
4	(1)	Sound Wave	17	(2)	Electric Circuit, Electrical Power
5	(4)	Self Inductance	18	(4)	Series LCR Circuit
6	(1)	Wave Optics	19	(2)	Atoms
7	(4)	Torque on a Current-carrying loop placed in a Magnetic field	20	(1)	Mechanical properties of Fluids
8	(3)	Motion in a Vertical Circle	21	[5.5]	Electronic Devices
9	(3)	Young's Double Slit Experiment	22	[205.00]	Thermodynamics
10	(3)	Logic Gates	23	[2.00]	Conservation of Mechanical energy
11	(3)	Moment of Inertia	24	[1.20]	Electrostatic Potential
12	(3)	Gravitation	25	[4.68]	Mechanical properties of Solids
13	(2)	Stopping Potential, X - Rays			

Chemistry

Q. No.	Answer	Topic's name	Q. No.	Answer	Topic's name
26	(1)	Periodic Table	39	(3)	Hydrocarbon
27	(4)	Chemical Bonding	40	(1)	Halogen Dervitative
28	(1)	Salt Analysis	41	(3)	P-block
29	(3)	Thermodynamics	42	(3)	Alcohol, Ether and Phenol
30	(2)	Ionic Equilibrium	43	(4)	Aldehyde and Ketone
31	(3)	Ionic Equilibrium	44	(1)	Carboxylic Acid
32	(1)	IUPAC	45	(3)	Nitrogen Compound
33	(1)	Isomerism	46	[15.00]	Chemical Kinetics
34	(1)	Hydrocarbon	47	[0.34]	Electrochemistry
35	(2)	Ketones	48	[2]	Alcohol
36	(2)	Lanthanoids	49	[57]	Thermodynamics
37	(3)	Solutions	50	[8]	Coordinaion Chemistry
38	(2)	General Organic Chemistry			